String Slicing

In [16]:
```
name='VijayaLaxmiReddy'
name
```

Out[16]:
```
'VijayaLaxmiReddy'
```

In [15]:
```
len(name)
```

Out[15]:
```
16
```

In [4]:
```
type(name)
```

Out[4]:
```
str
```

In [5]:
```
id(name)
```

Out[5]:
```
1792052410416
```

In [6]:
```
print(name)
```

```
VijayaLaxmireddy
```

In [9]:
```
name[:1]
```

Out[9]:
```
'V'
```

name[:]

In [11]:
```
name[2:7]
```

Out[11]:
```
'jayaL'
```

In [12]:
```
name[2:9:2]
```

Out[12]:
```
'jyLx'
```

In [17]:
```
int(2.3)
```

Out[17]:
```
2
```

Type Casting

INTEGER

In [18]:
```
int(2.5,5.6)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[18], line 1
----> 1 int(2.5,5.6)

TypeError: 'float' object cannot be interpreted as an integer
```

In [19]:
```
int('10')
```

Out[19]: 10

In [20]:
```
int(True)
```

Out[20]: 1

In [21]:
```
int(False)
```

Out[21]: 0

In [22]:
```
int(10+50j)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[22], line 1
----> 1 int(10+50j)

TypeError: int() argument must be a string, a bytes-like object or a real number,
not 'complex'
```

In [23]:
```
int('ten')
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[23], line 1
----> 1 int('ten')

ValueError: invalid literal for int() with base 10: 'ten'
```

FLOAT

In [24]:
```
float(80)
```

Out[24]: 80.0

In [25]:
```
float(True)
```

Out[25]: 1.0

In [26]:
```
float(10+30j)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[26], line 1
----> 1 float(10+30j)

TypeError: float() argument must be a string or a real number, not 'complex'
```

In [27]:
```
float('ishwari')
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[27], line 1
----> 1 float('ishwari')

ValueError: could not convert string to float: 'ishwari'
```

In [28]:
```
float('50')
```

Out[28]: 50.0

STRING

In [29]:
```
str(67)
```

Out[29]: '67'

In [30]:
```
str(True)
```

Out[30]: 'True'

In [31]:
```
str(20+50j)
```

Out[31]: '(20+50j)'

In [32]:
```
str(3.5)
```

Out[32]: '3.5'

BOOLEAN

In [33]:
```
bool(3.6)
```

Out[33]: True

In [34]:
```
bool(20+56j)
```

Out[34]: True

In [35]:
```
bool('45')
```

Out[35]: True

In [36]:
```
bool('6','8')
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[36], line 1
----> 1 bool(    ,    )

TypeError: bool expected at most 1 argument, got 2
```

COMPLEX

In [37]:
```
complex(True)
```

Out[37]: (1+0j)

In [38]:
```
complex(3.7)
```

Out[38]: (3.7+0j)

In [39]:
```
complex(67,90)
```

Out[39]: (67+90j)

In [40]:
```
complex('78')
```

Out[40]: (78+0j)

In [41]:
```
complex('8','90')
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[41], line 1
----> 1 complex(    ,     )

TypeError: complex() can't take second arg if first is a string
```

PYTHON BASICS OPERATORS

ARITHMETIC OPERATOR

In [42]:
```
x,y=9,8
print(x,y)
```

```
9 8
```

In [43]:
```
x-y
```

Out[43]: 1

In [44]:
```
x+y
```

Out[44]: 17

In [45]:
```
x*y
```

Out[45]: 72

In [46]:
```
x**y
```

Out[46]: 43046721

In [47]:
```
x/y
```

Out[47]: 1.125

In [48]:
```
x//y
```

Out[48]: 1

In [53]:
```
x%y
```

Out[53]: 0

ASSIGNMENT OPERATOR

In [57]:
```
x=60
x
```

Out[57]: 60

In [58]:
```
x=x+7
x
```

Out[58]: 67

In [61]:
```
x+=3
x
```

Out[61]: 65

In [62]:
```
x-=7
x
```

Out[62]: 58

In [64]:
```
x*=4
x
```

Out[64]: 928

In [65]:
```
x**=2
x
```

Out[65]: 861184

In [66]:
```
x/=8
x
```

Out[66]: 107648.0

In [67]:
```
x//=8
x
```

Out[67]: 13456.0

In [68]:
```
x%=9
x
```

Out[68]: 1.0

UNARY OPERATOR

In [69]:
```
n=6
n
```

Out[69]: 6

In [71]:
```
m=-n
m
```

Out[71]: -6

In [73]:
```
-n
```

Out[73]: -6

RELATIONAL OPERATOR

In [74]:
```
a=9
b=8
a,b
```

Out[74]: (9, 8)

In [75]:
```
a<b
```

Out[75]: False

In [76]:
```
a>b
```

Out[76]: True

In [78]:
```
a<=b
```

Out[78]: False

In [79]:
```
a>=b
```

Out[79]: True

In [80]:
```
a!=b
```

Out[80]: True

LOGICAL OPERATOR

In [81]:
```
a=7
b=3
a,b
```

Out[81]: (7, 3)

In [82]:
```
a<b and b<a
```

Out[82]: False

In [83]:
```
a>b and b>a
```

Out[83]: False

In [84]:
```
a<b and b>a
```

Out[84]: False

In [85]:
```
a>b and b<a
```

Out[85]: True

In [86]:
```
a>b or b<a
```

Out[86]: True

In [87]:
```
a<b or b>a
```

Out[87]: False

In [88]:
```
a<=b and b>=a
```

Out[88]: False

In [ ]:

In [ ]:

In [ ]: